



GROUNDWORK SF

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SF COMMUNITY SITE CONTEXT GRAPH

Every public-record claim carries its proof.

REACT · TYPESCRIPT · MAPLIBRE · CYTOSCAPE

THE RUNNING APPLICATION · DIGITALOCEAN AGENT, OPEN

DEMO SITES

300 De Haro Street

Showplace Square · 100% affordable ·
300 De Haro St

CONSTRUCTION

425 AFFORDABLE UNITS

1939 Market Street

Duboce Triangle · senior housing ·
1939 Market St

BP ISSUED

185 AFFORDABLE UNITS

758/772 Pacific Avenue

Chinatown · senior housing · 758/772
Pacific Ave

BP FILED

174 AFFORDABLE UNITS

FOCUS

OVERVIEW

HOUSING

PERMITS

HAZARDS

NEIGHBORHOOD



TRUST 100% CITED 7 SOURCES 15 ASSERTIONS 1 STALE 0 CONFLICTS 1 GAPS 1

DIAGNOSTICS

One graph, deployed on DigitalOcean.

Read path and agent path share one canonical graph – no site fact reaches the model except through a bounded, hash-verified packet.



The browser never receives a model key, database URL, or Spaces credential – only public API data and widget identifiers.

THE CONTRACT

The canonical graph model.

A typed application contract – not a model-generated structure.
One release nests every site, entity, and dated assertion
down to its evidence.



RELEASE · IMMUTABLE · SHA-256 MANIFEST

SITE · PARCEL IDENTITY · GEOMETRY



Entity

parcel · project · permit · assessment · program · hazard
· signal



Assertion

subject -predicate→ entity / literal · dated

↓ EVIDENCE_IDS[]



EvidenceRecord

dataset · record key · retrieved_at · source.updated_at · license · SHA-256



Diagnostic

freshness | conflict | coverage_gap | proximity_only



TrustSummary

citation coverage % + diagnostic counts

ENFORCED INVARIANTS

Every assertion references evidence. Missing rows stay **coverage gaps**; nearby records stay **proximity-only**;
conflicting dated assertions coexist – never silently reconciled.

Graph-aware RAG, hash-verified.



USER



WIDGET



GLM-5.2



FUNCTION



FASTAPI

**MANDATORY**

site facts via the Function;
methodology via the Knowledge Base

BOUNDED

#64 KiB · SHA-256 · one 5 s request ·
no redirects

REFUSES

valuation · legal · safety · ranking ·
eligibility · investment

DEPLOYED

Built on DigitalOcean.

Eight managed services, one project VPC – the complete path from public data to a grounded answer.

DIGITALOCEAN PROJECT - ONE REGION



App Platform

React + FastAPI container · public HTTPS · /healthz

PUBLIC EDGE

DATA PLANE · PRIVATE STORAGE



Managed PostgreSQL 17

snapshots · leases



Spaces

source + methodology buckets

AGENT PLANE



Agent Platform

GLN-S.2



Functions

Py 3.13 adapter



Knowledge Base

OpenSearch 2.19



Agent Evaluations fixed 50-query grounding · refusal · injection = promotion gate

packet = 65,536 B

SHA-256 verified

fixture fallback

/healthz · /api/data-status



DEEP DIVE ON THE ARCHITECTURE?

Message me – happy to walk through it.

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